

13th Week Summary (11/24/26)

- **Population proportion**

- Draw a large random sample of size n from a large population having unknown proportion p of successes.

To test the hypothesis $H_0 : \pi = \pi_0$, compute the Z statistic: $z = \frac{\hat{\pi} - \pi_0}{\sqrt{\frac{\pi_0(1 - \pi_0)}{n}}}$

- Use this test when the sample size n is so large that both $n\pi_0$ and $n(1 - \pi_0)$ are 5 or more.

- The p-value for a test of $H_0 : \pi = \pi_0$ against

$H_a : \pi > \pi_0$ is p-value = $P(Z \geq z)$

$H_a : \pi < \pi_0$ is p-value = $P(Z \leq z)$

$H_a : \pi \neq \pi_0$ is p-value = $2P(|Z| \geq |z|)$

- Confidence interval for π can be obtained by $\hat{\pi} \pm z_{\alpha/2} \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{n}}$. This interval should not be used unless $n\hat{\pi} \geq 5$ and $n(1 - \hat{\pi}) \geq 5$.

- **Chi-square Tests**

- The **chi-square goodness-of-fit** test allows us to test whether a categorical variable follows a given probability distribution.

- A categorical variable has k possible outcomes, with probabilities $\pi_1, \pi_2, \pi_3, \dots, \pi_k$. That is, π_i is the probability of the i^{th} outcome. We have n independent observations from this categorical variable.

- We use chi-square goodness-of-fit test for testing the null hypothesis that the probabilities have specified values:

$$H_0 : \pi_1 = \pi_{10}, \pi_2 = \pi_{20}, \pi_3 = \pi_{30}, \dots, \pi_k = \pi_{k0}$$

The expected count of outcome i , $E_i = n\pi_{i0}$. Expected counts do not have to be round numbers.

- The chi-square statistic is a measure of how far observed counts are from expected counts under the null hypothesis. The formula for the statistic is:

$$\chi^2 = \sum \frac{(\text{observed count} - \text{expected count})^2}{\text{expected count}} = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Each of the k terms in the sum is called a chi-square component

- The chi-square goodness of fit test involving k outcomes refers to the chi-square distribution with $k - 1$ degrees of freedom. The p-value is the area to the right of χ^2 under the density curve of this chi-square distribution.

- You can safely use the chi-square goodness-of-fit test with critical values from the chi-square distribution when no more than 20% of the expected counts are less than 5 and all individual expected counts are 1 or greater. In particular, the chi-square goodness-of-fit test can be used when all the expected counts are 5 or greater.

- The **chi-square test of independence** is for checking the dependence of two categorical variable. The null hypothesis is given by :

H_0 : Two categorical variables are independent of each other.

- The chi-square test for a two-way table with r rows and c columns uses critical values from the chi-square distribution with $(r - 1)(c - 1)$ degrees of freedom.

- You can safely use the chi-square test with critical values from the chi-square distribution when no more than 20% of the expected counts are less than 5 and all individual expected counts are 1 or greater. In the special case of a 2×2 table, all four expected counts should be 5 or greater.